

Deterministic geometric Savitzky-Golay dynamics estimator

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Shared parameter update: reference -> estimated
mass [kg]                2.351597591 -> 2.351597591    delta=+4.06774e-10    ratio=1
CoG x [m]                -0.002024708562 -> -0.002024708592    delta=-3.0192e-11
ratio=-
CoG y [m]                -3.052657894e-05 -> -3.052655096e-05    delta=+2.79791e-11
ratio=-
CoG z [m]                0.009509749599 -> 0.009509749623    delta=+2.31938e-11
ratio=-
rotor effectiveness 1    1 -> 0.9999999992    delta=-7.98834e-10    ratio=1
rotor effectiveness 2    1 -> 0.9999999996    delta=-4.10965e-10    ratio=1
rotor effectiveness 3    1 -> 0.9999999992    delta=-8.12559e-10    ratio=1
rotor effectiveness 4    1 -> 0.9999999989    delta=-1.1348e-09     ratio=1
command lag [s]          0 -> 0.0040000000253    delta=+0.004         ratio=-

Inertia matrix [kg m^2]
reference:
[ 0.06500006148, -7.27899253e-07, 1.901508003e-05]
[-7.27899253e-07, 0.06495265634, 5.9167305e-08]
[ 1.901508003e-05, 5.9167305e-08, 0.1289921107]
estimated:
[ 0.06500006149, -7.278984663e-07, 1.901508011e-05]
[-7.278984663e-07, 0.06495265636, 5.916766821e-08]
[ 1.901508011e-05, 5.916766821e-08, 0.1289921107]

Principal moments: reference -> estimated
principal moment 1 [kg m^2]    0.06495264516 -> 0.06495264518    delta=+1.54861e-11
ratio=1
principal moment 2 [kg m^2]    0.06500006701 -> 0.06500006702    delta=+8.27954e-12
ratio=1
principal moment 3 [kg m^2]    0.1289921163 -> 0.1289921163    delta=+7.97443e-12    ratio=1

Optimization
joint dynamics loss: 2.70871660784
Gaussian-prior cost: 1.89455249605e-16
selected strict-ZOH lag [s]: 0.0040000000253

Bag failure_1_single
reconstruction position RMSE [m]: 0.00455405797255
reconstruction position component RMSE [m]: [0.003382 0.001131 0.002832]
reconstruction orientation RMSE [deg]: 1.63617908637
gyro consistency RMSE xyz [rad/s]: [0.074577 0.140938 0.026029]
specific-force consistency RMSE xyz [m/s^2]: [0.851822 0.149748 0.274566]
inferred external force RMS xyz [N]: [2.039077 0.575899 3.95976 ]
inferred external torque RMS xyz [N m]: [0.124502 0.536087 0.103833]
bag dynamics loss: 2.70871660784

Recorded command timestamp intervals (data-derived)
Bag failure_1_single
rotor: median dt=0.00499773026 s (200.091 Hz), min=2.00271606e-05 s, max=0.0138893127 s
gimbal: median dt=0.00499868393 s (200.053 Hz), min=4.529953e-06 s, max=2.48282003 s
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